

Manasa Rapuru

Scientific Software Engineer

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<https://manasarapuru.github.io/scientific-software-website/>

SUMMARY

Engineer with a bioinformatics background, translating scientific needs into data-driven applications and AI-enabled tools. I bridge biology, software engineering, and user-centered design to build practical systems for scientific work.

EDUCATION & CERTIFICATIONS

M.S. Bioinformatics: Boston University	Sept 2021 – May 2023
B.S. Microbiology & Chemistry Minor: San Jose State University	Aug 2016 – Dec 2020
Artificial Intelligence in Pharma and Biotech Certification: MIT Management Sloan School	Aug 2026
AWS Cloud Practitioner Certification	Expected Oct 2026
IBM AI Developer Certification: Coursera	Jan 2026
IBM Full Stack Software Developer Specialization: Coursera	Apr 2024
Cancer Biology Specialization: Coursera	Jul 2026

RELEVANT EXPERTISE

Communication & Product Skills: Requirements Gathering, Stakeholder Collaboration, User-Centered Development, Workflow Design, Product Iteration, Interdisciplinary Communication, Cross-Disciplinary Collaboration

AI & Machine Learning: Prompt Engineering, Claude Code, Machine Learning, RAG, LLM Applications, AI agents

Computing: Python, Linux, R (DESeq2, ggplot2), SQL, Bash, Git, Java, Perl, Snakemake

Data Science: Pandas, NumPy, SciPy, Scikit-learn, Matplotlib

Cloud & Infrastructure: AWS, HPC/SLURM, CI/CD

Domain Knowledge: Genomics, transcriptomics, CRISPR, single-cell RNA-seq, Bulk RNA-Seq, NGS, molecular assay design, qPCR/dPCR

Software Engineering: React, REST APIs, Fast API, Flask, Streamlit, HTML, CSS, Javascript, Docker, AWS, Application Development, SDLC

Bioinformatics Tools: Scanpy, Seurat, BLAST, dbSNP, COSMIC, SAMtools, BEDTools, STAR, GATK, FastQC, Trimmomatic, GV, Ensembl, NCBI

RELEVANT WORK

UX Critic/Digital Story Teller Volunteer: California Academy of Life Sciences Aug 2026 - Present
Scientific Computing Department

- Using publicly available natural history archive databases to curate descriptions for academy's data
- Evaluating the user interface through interaction testing and reporting bugs

Member: US-Research Software Engineering Association Aug 2026 - Present
User Experience Working Group

- Learning how to gather and incorporate audience needs when developing scientific research software

Bioinformatics Scientist: Thermo Fisher Scientific Apr 2024 – Nov 2025
qPCR/dPCR TaqMan Assays

- Worked within a revenue-generating commercial qPCR/dPCR assay design service that involved designing for SNP-level allele discrimination, expression quantification, and multiplexing requirements.
- Collaborated with cross-functional scientific, engineering, and product stakeholders to translate complex assay development workflows into deployable computational tools
- Developed AI-enabled and software applications to streamline data workflows, reduce turnaround time and improve cross-team efficiency

Bioinformatics Specialist: Massachusetts General Hospital

Mar 2021 – Aug 2023

Huntington's Transcriptomics/Genomics

- Developed a Python-based analytical pipeline (Pandas, NumPy, Matplotlib, Scikit-Learn) to model cell morphology metrics derived from IncuCyte imaging.
- Evaluated machine learning approaches to assess whether custom models could improve phenotypic classification accuracy in comparison to the instrument's default analysis software.
- Integrated RNA-seq, single-cell RNA-seq, and tandem-repeat genomic analyses to investigate disease-associated gene expression patterns and molecular signatures in Huntington's disease, implementing differential expression workflows using R, DESeq2, and ggplot2.
- Designed reproducible computational workflows for HPC environments, automating data processing in Bash and integrating custom and precompiled bioinformatics tools within shared Linux infrastructure.

Bioinformatics Master's Student: Boston University

Sept 2021 – May 2023

Bioinformatics

- Designed and developed an interactive scientific data application enabling researchers to explore postmortem brain transcriptomic datasets across brain regions and disease conditions.
- Built the application using Python, SQL, JavaScript, HTML, and CSS, implementing interactive data querying and visualization for large-scale transcriptomic datasets.
- Developed reproducible Snakemake-based workflows for RNA-seq and single-cell RNA-seq analysis in R and Linux environments, integrating upstream computational pipelines with the interactive analytical application.

Bioinformatics Intern: Sandia National Laboratories

Jun 2020 – Aug 2020

Bioinformatics Research / Phage Therapy

- Applied computational genomics to support phage therapy research, using phylogenetic and comparative analyses to investigate integrative genetic elements in bacterial genomes.
- Worked within HPC-based computational environments to execute and optimize bioinformatics workflows, diagnosing infrastructure constraints affecting memory, I/O, and job scheduling.
- Collaborated on the development and improvement of Python and Perl bioinformatics pipelines, contributing changes to open-source repositories through Git.

Undergraduate Bioinformatics Research Student: San Jose State University

Jan 2019 – Dec 2020

Phylogenetics & Protein Modelling

- Investigated evolutionary relationships and potential functional roles of *Candidatus Saccharibacteria* within the oral microbiome through phylogenetic and protein visualization analyses, translating complex biological datasets into interpretable scientific insights.